Descriptive Statistics and Technical Interpretation

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1. Descriptive Statistics and Why They Matter

Descriptive statistics are numbers that summarize a dataset without requiring every individual value to be examined. Instead of scrolling through hundreds of voltage readings, a mean and a spread give an immediate sense of what the data looks like overall.

These summaries come before cleaning, visualization, feature creation, or machine learning because they give the first read on a dataset. A quick summary can flag a negative mass, a resistance in the millions, or a pH of 25 before anything is built on top of a broken column.

A summary only describes what the data looks like; it does not explain why it looks that way. A dataset can have a clean mean and a small standard deviation and still hide something important, because summarizing is not the same as understanding.

Example: pH readings pooled from two different labs — one consistently slightly acidic, the other slightly alkaline — can average out to almost exactly 7.0. The summary is technically correct and completely misleading, since it hides two very different groups of measurements sitting on either side of neutral.

2. Center and Spread

Center describes where the data tends to sit, and spread describes how much it wanders around that point. Two experiments can share the same mean and still behave very differently if one has a tight spread and the other is scattered widely.

- **Mean —** the sum of all values divided by the count; sensitive to extreme values.
- **Median —** the middle value once sorted; depends only on position, not magnitude.
- **Minimum / Maximum —** the smallest and largest recorded values.
- **Range —** the gap between maximum and minimum; driven by just two observations.
- **Variance —** how far values typically sit from the mean, in squared units.
- **Standard deviation —** the square root of variance, expressed in the original unit.
- **IQR (interquartile range) —** the spread of the middle 50% of the data ($Q3 - Q1$), ignoring the top and bottom quarters.

The mean becomes misleading when a few extreme values pull it away from where most of the data actually sits — five readings near 100 and one at 5,000 will drag the mean far above what a typical reading looks like. The median does not have this problem, since it depends only on order, not magnitude, so it stays close to the bulk of the data regardless of how extreme the outliers are.

A high standard deviation in an experiment usually means the readings are inconsistent — this could reflect a real physical effect, or it could mean the setup is unstable, the sensor is noisy, or conditions changed mid-run. The IQR holds up better than the range or standard deviation when data is skewed or contains outliers, because it deliberately ignores the extreme 25% on each end and describes only the ordinary middle portion of the data.

3. Distribution Shape and Sample Size

A distribution shows how values are spread across their possible range — whether they cluster near one point, spread evenly, or pile up at both ends — revealing more than a single average ever could. Skewness points to a lopsided dataset with a long tail on one side; clusters can indicate hidden subgroups, such as two machines producing slightly different results; and gaps can mean a measurement range that simply does not occur, or a threshold effect where values jump rather than vary smoothly.

A statistic from five readings is far weaker than one from five hundred, because a small sample is much more sensitive to random noise. One unusual reading in a batch of five can shift the mean drastically, while in five hundred readings the same odd value barely moves anything. This is why sample size matters before trusting a mean, a variance, or a pattern spotted in a plot.

Example: if a team tests five resistors from a new batch and finds one defective unit, the observed failure rate reads as 20%, even if the batch's true failure rate is closer to 2%. A single bad unit in a small sample can completely distort the conclusion.

4. Outliers and Domain Context

An outlier is a value that sits noticeably far from the rest of the data, standing apart from the general pattern everything else follows. Not every outlier is a mistake — some are genuine, rare events that happen to fall outside the normal range, and deleting them without checking throws away real information.

An outlier is worth investigating rather than deleting outright whenever there is a plausible explanation behind it, whether that is an equipment fault, a one-off environmental condition, or an actual rare occurrence in the system being measured. The value should be understood before it is removed.

- **Biochemistry:** a pH reading far outside the expected range could point to contamination, the wrong buffer solution, or a faulty probe.
- **Electronics:** a sudden current spike might mean an overload, a brief transient event, or an actual short circuit.
- **Mechanical:** an unusually high force reading could mean a genuinely stronger material sample, a unit mismatch, sensor saturation, or a real but rare event worth documenting.

5. Plot Selection and Interpretation

- **Histogram** — best for the distribution of one numeric variable, e.g. the spread of voltage readings across a batch of components.
- **Box plot** — best when center, spread, and outliers all need to be visible at once, e.g. tensile strength compared across three material batches.
- **Bar chart** — best for categorical data or counts, e.g. how many samples fall into each blood-type category.
- **Scatter plot** — best for the relationship between two numeric variables, e.g. current plotted against voltage.
- **Line plot** — best when values are ordered, usually across time, e.g. temperature logged every minute during an experiment.

Every plot needs a clear title, labelled axes, and units, because a plot without units is just shapes on a page — a reader has no way to know whether a spike represents 5 volts or 5,000 volts without that label.

Table 1: Plot Types and When to Use Them

Plot Type	Best Used For	Example
Histogram	Distribution of one numeric variable	Spread of voltage readings across a batch
Box Plot	Center, spread, and outliers, often across groups	Tensile strength across three material batches
Bar Chart	Counts across categories	Number of samples per blood type
Scatter Plot	Relationship between two numeric variables	Current plotted against voltage
Line Plot	Values ordered over time or sequence	Temperature logged every minute during a test

Table 1 above summarizes the five plot types described in this section and the situations each is best suited for.

6. Correlation, Causation, and Confounding

Correlation means two variables tend to move together, whether both are rising, both are falling, or one rises as the other falls. Causation means one variable's change actually brings about the change in the other — a much stronger and harder claim to prove than correlation.

Two variables increasing together does not prove that one caused the other, because the link could be coincidence, or both could be driven by some third factor that neither of them directly controls. That third factor is a confounding variable: something hidden that influences both variables at once, creating what looks like a direct relationship between them when there is not one.

Example: attendance and exam scores in a class often rise together. It is tempting to say attendance boosts scores directly, but student motivation could be the real driver behind both — a more motivated student shows up more and studies more, and the attendance–score link is really a byproduct of that shared cause.

7. Domain Measurement Interpretation and Comparability

Averaging pH, voltage, current, force, and pressure together produces a number with no real meaning, because these are entirely different physical and chemical quantities measured on different scales. Units are not just a label tacked onto a number — they are part of what the number means; a reading of 12 says nothing until it is known whether it refers to volts, newtons, or degrees Celsius.

Before any comparison makes sense, values need to be grouped by domain, measurement type, unit, and experimental condition. Comparing like with like is what makes a comparison valid in the first place.

Valid comparisons:

- Average voltage under low-load versus high-load conditions, using the same measurement setup and unit.
- pH of a solution before and after a treatment step, using the same probe and scale.

- Tensile strength across two batches of the same material, measured in the same unit under the same test method.

Invalid or misleading comparisons:

- Comparing pH directly against voltage, since they measure entirely different things.
- Comparing force values where one set is in Newtons and the other in pounds-force without converting first.
- Averaging results from two unrelated experiments just because both happen to produce numbers.

Before analyzing domain measurement data, it is worth asking for the exact unit used, the conditions under which the data was collected, how the instrument was calibrated, the sample size, and how missing or invalid readings were handled.

8. Final Reflection

How can statistics be technically correct but still scientifically or practically misleading?

Statistics can be arithmetically correct while still being scientifically or practically misleading whenever the number is separated from the context that produced it. A mean is only as meaningful as the unit attached to it; a value of 42 says nothing until it is known whether it represents volts, newtons, or degrees. Mixing different measurement types, such as averaging pH with voltage, produces a number with no real-world interpretation even though the calculation itself is valid. Small sample sizes make the problem worse, because a single unusual reading can swing a mean or standard deviation far more than it should, while outliers add another layer of risk: removing them blindly can erase a genuine finding, and ignoring them can hide a real problem. Correlation is frequently mistaken for causation, especially when a hidden confounding variable is driving both quantities at once. Every statistic also carries assumptions about how it was measured, calibrated, and recorded. A number is only trustworthy once its unit, sample size, and experimental context have been checked.

Appendix: Review Question Answers

1.

It suggests the data is skewed or contains outliers pulling the mean away from where most values actually sit.

2.

A standard deviation of 5 means something completely different for millimeters than for kilometers — without knowing what was measured and under what conditions, the number cannot be judged as large or small.

3.

No. A sudden spike in force during a stress test could reveal a genuine material failure threshold, which is valuable information rather than an error to discard.

4.

It is still a number, so it fits the numerical data type, but percentages are capped at 100 by definition, so a value above that points to a data entry or calculation error rather than a type mismatch.

5.

These quantities are measured on different scales for different physical properties, and combining them mathematically does not correspond to anything that exists physically.

6.

The median, because it only depends on the position of values once sorted, not their actual magnitude, so an extreme value does not drag it away from the bulk of the data.

7.

A box plot shows the spread, median, and outliers of numeric data in one view, while a bar chart shows only counts or totals per category, with no sense of spread within a category.

8.

It can be concluded that the two move together across the group observed. It cannot be concluded that attending more classes directly causes higher scores, since a shared factor such as motivation could be driving both.

9.

"The average temperature across the lab and the workshop today was 24°C" is arithmetically correct but useless, since the two rooms are unrelated environments and averaging them describes neither one accurately.

10.

The unit of measurement, the instrument used and its calibration status, the experimental conditions, the sample size, and how any missing or invalid readings were handled.

References

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